



Fig. 2: Semiconductors exhibit an increase in mobility at low temperature

Transition Temperature and Critical Magnetic Field of Some Superconducting Elements

Superconducting elements	Transition temperature T _c (degree Kelvin)	Critical magnetic field H _c (oersted)
Zinc	0.88	53
Aluminium	1.2	99
Indium	3.41	282
Tin	3.72	306
Mercury	4.15	411
Lead	7.19	803

Some of the materials used in cryoelectronics

- Semiconductors like silicon, germanium and gallium arsenide
- CMOS
- High-temperature co-fired silicon
- Amorphous silicon
- Superconductors

different when cooled to exceedingly low temperatures. These become superconductors at such temperatures.

Superconductivity was discovered in 1911 by Dutch physicist Heike Kamerlingh Onnes, who was awarded Nobel Prize in 1913 for his low temperature research.

Hundreds of materials are known to become superconducting at low temperatures. Twenty six of the chemical elements (all of them metals) are superconductors in their usual crystallographic forms at at-

mospheric pressure. Aluminium, tin, lead and mercury are a few among these. In addition, thirteen chemical elements that are metals, semi-metals and semiconductors act as superconductors at low temperatures and high pressures. Among these are uranium, silicon and selenium.

Semiconductors exhibit an increase in mobility at low temperatures (shown in Fig. 2). This is because at low temperature their atomic lattice vibration decreases and mobility increases significantly.

The best superconductors today are alloys or compounds. Examples include disilver fluoride or silver subfluoride (Ag₂F) and compound of carbon and potassium (C₆K). Some semiconducting compounds, such as tin telluride (SnTe), can become superconducting if properly infused with impurities.

Transition temperature and critical magnetic field of some superconducting elements are shown in the table.

Cryoelectronics: milestone in space technology

NASA is using components such as semiconductor switching devices, magnetic capacitors, digital-to-analogue and analogue-to-digital converters, DC-AC converters, operational amplifiers, oscillators, and power conversion and conditioning circuits based on cryoelectronics. Cryoelectronics is also used in the driver circuitry of motors or actuators. Commercial high-speed pulse-width-modulation (PMW) chips have been characterised in terms of their performance in the range of 25°C to -196°C. Silicon-

on-insulator can be used to make MOS devices with gates as small as 40 nm.

Advantages of cryoelectronics for space technology are increased circuit speed, low power dissipation, fast switching, high semiconductor and metal thermal conductivity, reduction in thermally induced failures of components and devices, increased integration density primarily because of reduced operating supply voltages, and improved digital and analogue circuit performance (in terms of switching speed, noise margin and gain bandwidth).

Disadvantages include novel fabrication technique required, low carrier mobility, high cost of circuit and high volume.

Cryoelectronics finds applications in outer-space satellites, e.g., in sensors for flow, temperature and pressure measurement; detectors of cosmic radiation, such as infrared and ultra-violet detectors; nuclear astronomy instruments; medical instruments; dynamic random-access memories; high-end computers and organic electronics.

Latest developments

The possibility to confine light to the nanoscale and researches in superconductivity have evoked a large interest, leading to rapid growth of cryoelectronics. Cryogenic robotics, low-temperature electronics and high-temperature superconductivity need to be collaborated to shape the future of lunar research and exploration.

One can imagine a future where heat-source-free cryoelectronics technologies can be tapped for sensors and other electronic components of satellites, ultra-low-temperature freezers, cryogenic surgical instruments and computers. ●

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